# Switch and Router Configuration Theory

**Task 1: Create VLAN 10 (Sales) and VLAN 20 (Support) on the Switch**

1. Open **Cisco Packet Tracer**, drag a **2960 Catalyst Switch** onto the workspace, and open its **CLI** tab.
2. Enter the configuration commands:

Code snippet

Switch> enable

Switch# configure terminal

! Create VLAN 10 and name it Sales

Switch(config)# vlan 10

Switch(config-vlan)# name Sales

Switch(config-vlan)# exit

! Create VLAN 20 and name it Support

Switch(config)# vlan 20

Switch(config-vlan)# name Support

Switch(config-vlan)# exit

**Task 2: Assign Switch Ports to VLANs & Verify with show vlan**

**1. Port Assignment Commands (Using interface range):**

- Assign 3 ports (Fa0/1 to Fa0/3) to **VLAN 10 (Sales)**:

Code snippet

```
Switch(config)# interface range fa0/1 - 3
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit
```

- Assign 2 ports (Fa0/4 to Fa0/5) to **VLAN 20 (Support)**:

Code snippet

```
Switch(config)# interface range fa0/4 - 5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# end
Switch# write memory
```

**2. Verification Command:**

Code snippet

```
Switch# show vlan brief
```

**Expected Command Output:**

Plaintext

```
VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/6, Fa0/7, Fa0/8, Fa0/9...
10   Sales                             active    Fa0/1, Fa0/2, Fa0/3
20   Support                           active    Fa0/4, Fa0/5
```

- **Ports Fa0/1, Fa0/2, and Fa0/3** are now active members of **VLAN 10**.
- **Ports Fa0/4 and Fa0/5** are members of **VLAN 20**.

**Task 3: Connect Two Routers and Configure a Static Route on Router 1**

**Network Topology Setup:**

- **Router 1 (R1):**
    - LAN Interface (g0/0): 192.168.10.1 /24
    - WAN Serial Interface (s0/1/0): 10.0.0.1 /30 (Clock Rate DCE)
- **Router 2 (R2):**
    - WAN Serial Interface (s0/1/0): 10.0.0.2 /30
    - LAN Interface (g0/0): 192.168.20.1 /24 (Target LAN behind R2)

**1. Router 1 Interface Configuration:**

Code snippet

Router1> enable

Router1# configure terminal

Router1(config)# hostname R1

! Configure WAN Link

R1(config)# interface serial 0/1/0

R1(config-if)# ip address 10.0.0.1 255.255.255.252

R1(config-if)# clock rate 64000

R1(config-if)# no shutdown

R1(config-if)# exit

! Configure Local LAN

R1(config)# interface gigabitethernet 0/0

R1(config-if)# ip address 192.168.10.1 255.255.255.0

R1(config-if)# no shutdown

R1(config-if)# exit

**2. Router 2 Interface Configuration:**

Code snippet

Router2> enable

Router2# configure terminal

Router2(config)# hostname R2

! Configure WAN Link

R2(config)# interface serial 0/1/0

R2(config-if)# ip address 10.0.0.2 255.255.255.252

R2(config-if)# no shutdown

R2(config-if)# exit

! Configure LAN behind R2

R2(config)# interface gigabitethernet 0/0

R2(config-if)# ip address 192.168.20.1 255.255.255.0

R2(config-if)# no shutdown

R2(config-if)# exit

**3. Static Route Configuration on Router 1:**

To tell Router 1 how to reach the 192.168.20.0/24 network behind Router 2:

Code snippet

R1(config)# ip route 192.168.20.0 255.255.255.0 10.0.0.2

R1(config)# end

R1# write memory

- **Command Syntax:** ip route [Destination Network] [Subnet Mask] [Next-Hop IP]
- 192.168.20.0: The destination subnet behind Router 2.
- 255.255.255.0: The subnet mask.
- 10.0.0.2: The next-hop IP address on Router 2's serial interface.

**Task 4: Why Sales-to-Support Ping Fails & Inter-VLAN Solutions**

**1. Why the Ping Fails:**

- **Layer 2 Broadcast Domain Isolation:** A VLAN splits a single physical switch into separate virtual switches. By default, a switch will never forward frames between different VLAN IDs.
- **Different Subnets:** Devices in Sales belong to 192.168.10.0/24, while Support devices belong to 192.168.20.0/24. A standard Layer 2 switch operates only on MAC addresses and cannot inspect or route IP packets.
- **Result:** When PC-Sales (192.168.10.10) pings PC-Support (192.168.20.10), the packet is dropped immediately, resulting in **"Request timed out"**.

**2. Solution: Enabling Inter-VLAN Communication**

**Option A: Router-on-a-Stick (ROAS)**

Connect one physical cable between the switch and a router, then configure sub-interfaces with 802.1Q802.1\text{Q}802.1Q trunking:

1. **Switch Trunk Port:**

Code snippet

Switch(config)# interface gigabitethernet 0/1

Switch(config-if)# switchport mode trunk

2. **Router Sub-Interfaces:**

Code snippet

Router(config)# interface g0/0.10

Router(config-subif)# encapsulation dot1Q 10

Router(config-subif)# ip address 192.168.10.1 255.255.255.0

Router(config)# interface g0/0.20

Router(config-subif)# encapsulation dot1Q 20

Router(config-subif)# ip address 192.168.20.1 255.255.255.0

**Option B: Layer 3 (Multilayer) Switch**

Use Switched Virtual Interfaces (SVIs) directly on a Layer 3 switch (e.g., Cisco 3560/3650):

Code snippet

Switch(config)# ip routing

Switch(config)# interface vlan 10

Switch(config-if)# ip address 192.168.10.1 255.255.255.0

Switch(config)# interface vlan 20

Switch(config-if)# ip address 192.168.20.1 255.255.255.0

**Complete Architecture & Configuration Visual Reference**

A reference architecture card illustrating both the switch VLAN allocations and the router static route: